

Class - 7

25/9/23

Ch - 3

Time Value of Money

Knowledge based

1. What is inflation (pg - 3/207)

2. What is opportunity cost?

3. Compounding Vs Discounting.

Future :- Tk. 50,000 (lump sum amount)

→ Simple interest :-

Principal

50,000 × 10% × 1 = 5,000

→ Compounding Interest :-

1st year :- Principal

2nd year :- 51,000

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Class-8

12/10/25

Ch-13

Time Value of Money

▣ Compounding :- Present value $\rightarrow$ Future

Value

▣ Discounting :- Future $\rightarrow$ Present

Principal + Interest

1. Simple Interest

2. Compounding

▣ Simple Interest :-

Principal

10% simple interest on ₹. 1,000

▣ Compounding Interest :-

1st year :- Principal

2nd year :- $P + i$

Annuity

- 1) Fixed Interval (5 years)
- 2) Same amount of money
- 3) Beginning or ending.

Types of Annuity

- 1) Ordinary Annuity
- 2) Annuity Due
- 3) Perpetuity
- 4) Deferred Annuity

cp

Class - 9

14/10/25

Ch-3 (Time Value of Money)

Mixed Stream

Single Stream:-

$$FV = PV \left(1 + \frac{i}{n}\right)^{nm}$$

$$PV = \frac{FV}{\left(1 + \frac{i}{n}\right)^{nm}}$$

cw//

Class - 10

18/5/25

Ch. 3: (Time Value of Money)

Lump Sum, $PV = \frac{FV}{(1 + \frac{i}{m})^{nm}}$

$$FV = PV \left(1 + \frac{i}{m}\right)^{nm}$$

* If you want to get Tk. 5,000 after 5 years at 12% monthly interest. How much money you have to invest today?

Here,

$$FV = \text{Tk. } 5,000$$

$$i = 12\% \text{ or } 0.12$$

$$n = 5$$

$$m = 12$$

We know,

$$PV = \frac{FV}{\left(1 + \frac{i}{m}\right)^{nm}}$$

$$= \frac{5,000}{\left(1 + \frac{0.12}{12}\right)^{12 \times 5}}$$

$$= \text{Tk. } 2,752.245$$

* Mr. Hasan deposits Tk. 2,000 at 10% interest twice in a month. How much money will he get in 3 years?

Sol:

$$FV = PV \left(1 + \frac{i}{m}\right)^{nm}$$

$$= 2,000 \left(1 + \frac{0.1}{24}\right)^{3 \times 24}$$

$$= \text{Tk. } 2,698.035$$

QW

Class - 11

16/10/25

Ch. 13

Annuity

* Present Value

→ Ordinary → ending

$$L \rightarrow PV_{OA} = A \times \left\{ \frac{1 - \frac{1}{(1 + \frac{i}{m})^{nm}}}{\frac{i}{m}} \right\}$$

→ Annuity ^{due} → beginning

$$L \rightarrow PV_{AD} = A \times \left\{ \frac{1 - \frac{1}{(1 + \frac{i}{m})^{nm}}}{\frac{i}{m}} \right\} (1 + \frac{i}{m})$$

* Future Value

→ Ordinary → ending

$$L \rightarrow FV_{OA} = A \left\{ \frac{(1 + \frac{i}{m})^{nm} - 1}{\frac{i}{m}} \right\}$$

→ Annuity Due → beginning

$$L \rightarrow FV_{AD} = A \left\{ \frac{(1 + \frac{i}{m})^{nm} - 1}{\frac{i}{m}} \right\} (1 + \frac{i}{m})$$

Mr. Rakib wants to save a certain amount of money each year to purchase a laptop worth Tk. 1,00,000. after 5 years. Determine how much money Mr. Rakib needs to deposit in the bank @ 10% interest.

Sol:

~~$$PV_{OA} = A \times \left\{ 1 - \frac{1}{(1+i)^n} \right\}$$~~

~~$$\text{or, } 1,00,000 = A \times \left\{ 1 - \frac{1}{(1+0.1)^5} \right\}$$~~

~~$$FV_{OA} = A \times \left\{ \frac{(1+i)^n - 1}{i} \right\}$$~~

~~$$\text{or, } 1,00,000 = A \times \left\{ \frac{(1+0.1)^5 - 1}{0.1} \right\}$$~~

~~$$\text{or, } 1,00,000 = A \times 6.1051$$~~

$$\text{or, } A = \frac{1,00,000}{6.1051}$$

$$\therefore A = \text{Tk. } 16,379.75$$

* Mr. Mahmudul opened a special savings account with a maturity period of 5 years. Monthly deposit is Tk. 2,000. The bank will pay 0.5% interest. How much money will he get after 5 years?

Sol:

$$FV_{OA} = A \times \left\{ \frac{(1 + \frac{i}{m})^{nm} - 1}{\frac{i}{m}} \right\} (1 + \frac{i}{m})$$

$$= 2,000 \times \left\{ \frac{(1 + \frac{0.005}{12})^{12 \times 5} - 1}{\frac{0.005}{12}} \right\} (1 + \frac{0.005}{12})$$

~~$$= 2,000 \times 202.7380 \times 1.00701$$~~

$$= \text{Tk. } 1,52,844.15$$

Ques

Ques 14

23/10/25

Time Value of Money

Mixed stream of cashflow :-

* Finding

$$FV = PV_1(1+i)^{n-1} + PV_2(1+i)^{n-2} + PV_3(1+i)^{n-3} + \dots + PV_n(1+i)^{n-n}$$

* Beginning

$$FV = PV_1(1+i)^n + PV_2(1+i)^{n-1} + PV_3(1+i)^{n-2} + \dots + PV_n(1+i)^{n-n+1}$$

* Mr. Tanvin deposit Tk. 1,000 at the end of 1st year, Tk. 2,000 at the end of 2nd year, Tk. 3,000 at the end of 3rd year and Tk. 1,500 at the end of 4th year at 10%.. what will be the FV of his deposited money?

Sol:

we know,

$$FV = PV_1(1+i)^{n-1} + PV_2(1+i)^{n-2} + PV_3(1+i)^{n-3} + PV_4(1+i)^{n-4}$$

$$= 1,000(1+0.10)^{4-1} + 2,000(1+0.1)^{4-2} + 3,000(1+0.1)^{4-3} + 4,000(1+0.1)^0$$

$$= 1,000(1.1)^3 + 2,000(1.1)^2 + 3,000(1.1)^1 + 4,000(1)$$

$$= 1,000(1.331) + 2,000(1.21) + 3,000(1.1) + 4,000$$

$$= 1,331 + 2,420 + 3,300 + 4,000$$

$$= 11,051$$

QW

Class - 15

26/10/25

Ch. 8

Time Value of Money

*Razzak is a student. His grand father wants to give him some money. So, he proposed 3 alternatives:-

Alt: 1 - Tk. 2500/2800/2200/2500/4800 in the next 5 years.

Alt: 2 - Tk. 2000/2800/2200/2500/4800 at the beginning of next 5 years.

Alt: 3 - Tk. 3000 at the end of each year.

Opportunity cost ~~5%~~ 8%.

c) P.V of Alt: 1.

d) Better Alt.

1st Alternative

a

Investment with time

$$\begin{aligned} PV &= \frac{FV_1}{(1+i)^1} + \frac{FV_2}{(1+i)^2} + \frac{FV_3}{(1+i)^3} + \frac{FV_4}{(1+i)^4} + \frac{FV_5}{(1+i)^5} \\ &= \frac{2500}{(1+0.08)^1} + \frac{2800}{(1+0.08)^2} + \frac{2200}{(1+0.08)^3} + \frac{2500}{(1+0.08)^4} + \frac{4800}{(1+0.08)^5} \\ &= \text{Tk. } 11,566.16842 \end{aligned}$$

∴ PV of 1st Alt = 11,566.16842 Tk,

(Am)

11,566.16842 Tk

d

2nd Alternative

$$\begin{aligned} PV &= \frac{FV_1}{(1+i)^0} + \frac{FV_2}{(1+i)^1} + \frac{FV_3}{(1+i)^2} + \frac{FV_4}{(1+i)^3} + \frac{FV_5}{(1+i)^4} \\ &= \frac{2500}{(1+0.08)^0} + \frac{2800}{(1+0.08)^1} + \frac{2200}{(1+0.08)^2} + \frac{2500}{(1+0.08)^3} + \frac{4800}{(1+0.08)^4} \\ &= \text{Tk. } 12,491.46189 \end{aligned}$$

3rd Alternative:

$$PV_{OA} = A \times \left[\frac{1 - \frac{1}{(1+i)^n}}{i} \right]$$

$$= Tk. 3,000 \times \left[\frac{1 - \frac{1}{(1+0.08)^5}}{0.08} \right]$$

$$= Tk. 3,000 \times 2.002710037$$

$$= Tk. 11,978.13014.$$

QW

Class - 16

26/10/25

Ch-3

Time Value of Money

* Mr. Shakin got Tk. 10,00,000 after selling his land. He wants to invest the money in 'M' Bank yielding Tk. 12,00,000 after 5 years. On the other hand, 'N' Bank proposed him Tk. 14,30,000 after 8 years. Which bank should he choose

Solⁿ:

For 'M' Bank,

we know,

$$FV = PV (1+i)^n$$

$$\text{or, } 12,00,000 = 10,00,000 (1+i)^5$$

$$\text{or, } \frac{12,00,000}{10,00,000} = (1+i)^5$$

$$\text{or, } \sqrt[5]{1.2} = \sqrt[5]{(1+i)^5}$$

$$\text{or, } (1.2)^{\frac{1}{5}} = (1+i)$$

$$\text{or, } 1.03713 = 1+i$$

$$\text{or, } i = 1.03713 - 1$$

$$= 0.03713 \times 100$$
$$= 3.71 \%$$

for 'N' Bank,

We know,

$$FV = PV(1+i)^n$$

$$\text{or, } 14,50,000 = 10,00,000(1+i)^8$$

$$\text{or, } \frac{14,50,000}{10,00,000} = (1+i)^8$$

$$\text{or, } \sqrt[8]{1.45} = \sqrt[8]{(1+i)^8}$$

$$\text{or, } 1.0475 = 1+i$$

$$\text{or, } i = 1.0475 - 1$$

$$\text{or, } i = 0.0475 \times 100$$

$$\therefore i = 4.75\%$$

$\therefore$ He should choose 'N' Bank.

is deposited

*If a certain amount of money at an interest rate of 12%, In how many years will it triple?

Solⁿ:

Let,

$$PV = Tk. 1,00,000$$

$$i = 12\% \text{ or } 0.12$$

$$FV = 1,00,000 \times 3 = Tk. 3,00,000$$

We know,

$$FV = PV(1+i)^n$$

$$\text{or, } 3,00,000 = 1,00,000(1+0.12)^n$$

$$\text{or, } \frac{3,00,000}{1,00,000} = (1+0.12)^n$$

$$\text{or, } 3 = (1+0.12)^n$$

$$\text{or, } \log 3 = n \log 1.12$$

$$\text{or, } n = \frac{\log 3}{\log 1.12}$$

$$\therefore n = 9.7 \text{ years. (Ans.)}$$

Examples of

no to position to income statement to 92%

*m = continuous compounding

given and all the other interest

*EIR = $e^i - 1$ convert to this way

*FV = $PV e^{inx}$

1000

*PV = $\frac{FV}{e^{inx}}$

1000

100,000, 1.1 = PV

100,000, 1.1 = i

100,000, 1.1 = PV

100,000

FV = $PV(1+i)^n$

$(100,000 + 100,000) = 200,000$

$(100,000) = \frac{200,000}{1.1}$

$(100,000) = 181,818$

100,000 = 181,818

$\frac{181,818}{1.1} = 165,289$

(100,000) = 165,289

Q10

class - 17

28/10/25

Pg - 200

Ex - 3.29

$$PV_{OA} = A \times \left[\frac{1 - \frac{1}{(1+i)^n}}{i} \right]$$

$$= 12,000 \times \left[\frac{1 - \frac{1}{(1+0.09)^{25}}}{0.09} \right]$$

$$= 12,000 \times 9.82257$$

$$= \text{Tk. } 1,17,870.84$$

Again,

(Ans.)

$$PV_{OA} = A \times \left[\frac{1 - \frac{1}{(1+i)^{n \times m}}}{\frac{i}{m}} \right]$$

$$= 1,200 \times \left[\frac{1 - \frac{1}{(1+\frac{0.09}{12})^{12 \times 25}}}{\frac{0.09}{12}} \right]$$

$$= 1,200 \times 119.16162$$

$$= \text{Tk. } 1,42,993.944$$

(Ans.)

Ex-203.30

Here,

$$\text{Tk. } (50\% \text{ of } 50 \text{ lakh}) = 25 \text{ lakh}$$

= Tk. 25 lakh = Remaining amount of installment

$$PV_{OA} = A \times \left[\frac{1 - \frac{1}{(1+i)^n}}{i} \right]$$

$$= \frac{4,40,000}{25,00,000} \times \left[\frac{1 - \frac{1}{(1+0.12)^{10}}}{0.12} \right]$$

$$= \text{Tk. } 25,00,000 \times 5.65022$$

$$= \text{Tk. } 24,86,096.8$$

∴ Payment of second option

$$= \text{Tk. } (24,86,096.8 + 25,00,000)$$

$$= \text{Tk. } 44,86,096.8$$

∴ He should choose 2nd option.

Ex- 3.34

~~$$PV_{OA} = 25,000 \times \left[1 - \frac{1}{(1+0.08)^n} \right]$$~~

Hence,

$$A = \text{tk. } 25,000$$

$$i = 8\% \text{ or } 0.08$$

we know,

$$PV_p = \frac{25,000}{0.08}$$

$$= \text{tk. } 3,12,500$$

(Ans.)

ew

Class - 10

MHK Sin
02/11/25Time Value

* Mr. Jamil wants to purchase a flat after 5 years from today. At that time, he has to pay Tk. 50,00,000. In this regard, he wants to deposit equal amount of money in Janata Bank/monthly $i = 6.5\%$.

c) $A = ?$

Sol:

$$FV_{OA} = A \times \left[\frac{\left(1 + \frac{i}{m}\right)^{n \times m} - 1}{\frac{i}{m}} \right] \left(\frac{1}{1 + \frac{i}{m}} \right)$$

$$\text{or, } 50,00,000 = A \times \left[\frac{\left(1 + \frac{0.065}{12}\right)^{12 \times 5} - 1}{\frac{0.065}{12}} \right] \left(\frac{1}{1 + \frac{0.065}{12}} \right)$$

$$\text{or, } \frac{50,00,000}{70.67396} = A$$

$$\therefore A = \text{Tk. } 70,747.41531$$

* Mr. Hasan taken a loan of Tk. 5,00,000 from Rupsha Bank at 10% for 6 years. He has to pay this loan by paying equal installment for the next 6 years. What is the amount of annuity?

Sol:

$$PV_{OA} = A \times \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

$$Tk. 5,00,000 = A \times \left[\frac{1 - (1+0.10)^{-6}}{0.10} \right]$$

$$Tk. \frac{5,00,000}{4.4859} = A$$

$$\therefore A = 1,11,459 \text{ Tk.}$$

CW 11 class - 22

11/11/25

ch. 3 (1st Paper)

* If you deposit Tk. 10,000 at 10%
continuous compounding, what will be
the EAR?

Soln:

$$\text{EAR} = e^i - 1$$
$$= e^{0.10} - 1$$

$$= 10.5\%$$